Prior-Robust QSVT Filter Selection for Gap-Free Spectral-Mean Estimation

Abstract

Selecting a spectral filter by its conditional convergence alone ignores the cost of rejected preparations. We study a specified alternative objective: minimizing the worst expected query cost of a filtered-mean estimator when only an interval containing the smallest normalized Gram eigenvalue and a lower bound on its initial probability mass are supplied. No spectral-gap bound or true target energy enters the design. A two-point characterization turns the uniform mean-bias requirement into coupled amplitude constraints. Our main structural result shows that imposing a zero at the upper spectral endpoint costs only one additional Gram-polynomial degree and a controlled factor in worst-case acceptance. In this class, a semi-infinite linear program exactly characterizes the best worst-case restart probability at each degree. We combine rational dual bounds, continuous Bernstein certificates and phase-response enclosures to select the degree and error allocation with a posteriori cost guarantees. Numerical experiments compare optimized, power, cosine-Taylor and Chebyshev responses under a common signal-query model. The results identify preparation-cost-dependent choices and quantify their distance from the restricted optimum. The contribution is a structural and certified design analysis of a fixed estimator and restart model, not a new QSVT primitive or a general quantum speedup.

1 Introduction

Quantum singular-value transformation (QSVT) implements polynomial responses of a block-encoded matrix [1, 2]. This flexibility motivates choosing different functions for different spectral problems. Such a choice is useful only when the input information used to select it is available and the preparation, normalization, readout and rejection costs are counted. A low-degree filter with a small accepted-state error can still be an expensive estimator if its acceptance probability is small.

We make this choice precise for one deliberately restricted, verifiable setting: a nonadaptive real polynomial filter followed by independent Bernoulli measurements of the normalized right Gram operator. The minimum lies in a supplied interval, its initial mass is lower bounded, and no gap is promised. The design must work for every finite spectrum compatible with this information, rather than selecting a filter after examining the true eigenvalues. The estimator is not intended to replace phase estimation or modern ground-energy estimators in general.

Three results organize the analysis. First, the robust mean-bias condition is equivalent to a two-energy inequality. Second, multiplying an admissible filter by $1 - x$ preserves this condition and gives an explicit acceptance comparison with an endpoint-nulling filter. Third, endpoint nulling makes the worst acceptance exactly the minimum target transmission times the overlap bound. This identifies a semi-infinite linear program whose objective directly controls worst expected restart cost. Exact continuous feasibility certificates and corrected dual bounds then turn numerical design into a quantitatively certified selection rule.

Relation to existing filtering. Powers of cosine, spectral-ratio bounds, shifts and amplification already appear in ground-state preparation [3]. Optimal polynomial projectors with a known isolated target are established [4]. Cosine and Gaussian filtering and virtual filtered observables have also been studied [5, 6]. Filtered quantum phase estimation explicitly balances filter cost and overlap, considers different bases, and gives known-center minimax constructions [7]. Wall-Chebyshev work analyzes robustness to coarse energy information together with success costs [8]. Recent constrained QSP minimax work addresses the distinction between sampled and continuous feasibility and connects optimization to phase synthesis [9]. Linear programming, minimax design, Bernstein subdivision and QSP synthesis are therefore background tools [10, 11, 12], not claimed inventions here.

Our narrower contribution is the endpoint-nulling reduction with its acceptance factor, the exact restart interpretation of the unknown-minimum coupled constraints, and certified selections for this particular gap-free mean estimator. Unlike a fixed-center rejection-band objective, our constraint couples every possible target in the prior interval to every larger possible eigenvalue. Unlike fidelity guarantees, it allows nearby unwanted eigenvalues to remain. These differences define the task; they do not establish superiority over existing estimators or universal priority for all the component arguments.

2 Information, oracle and estimator model

Let $B = A/\alpha$, where a known $\alpha \geq \|A\|_2 > 0$ is part of the input, and $G = B^\dagger B$ has eigenvalues in $[0, 1]$. Let $e = \lambda_{\min}(G)$. The supplied information is

$$0 \leq e \leq u < 1, \qquad p_0 \geq a, \quad 0 < a < 1, \tag{1}$$

where p_0 is total initial probability on the minimum eigenspace. All other eigenvalues may be arbitrarily close to e. Degeneracy is included by combining weights in the target eigenspace. There is no assumption that the spectrum or its weights can be read classically. The numerical generator knows them only to audit the resulting guarantees.

A real amplitude P obeys $|P(x)| \leq 1$ on $[0, 1]$ and is nonzero on $[0, u]$. For polynomials we choose its sign so $P > 0$ there. The accepted state has weights proportional to $p_i P(\lambda_i)^2$, success $Z_P = \sum_i p_i P(\lambda_i)^2$, and mean

$$\mu_P = \frac{\sum_i p_i P(\lambda_i)^2 \lambda_i}{Z_P}.$$

For robustness let $\mathfrak{I}(a, u)$ denote all such finite spectra and weights, and define

$$S(P) = \inf_{I \in \mathfrak{I}(a,u)} Z_P(I), \qquad P \text{ admissible at } b \iff \mu_P(I) - e(I) \leq b \quad \forall I.$$

Success is physically implemented with amplitude κP, for the common margin $\kappa = 19/20$, applied once to the entire polynomial. This changes success by κ^2 and leaves conditional means unchanged.

For a degree-n polynomial in $x = \lambda$, $P(s^2)$ is an even degree-$2n$ polynomial in the normalized singular value s. We assume an ideal controlled singular-value signal and its inverse, with the standard QSVT reflections available. A conservative serial implementation of real-valued response extraction uses a controlled QSP unitary and its inverse: $4n$ signal calls per complete filter attempt. This convention is used for every candidate. It includes neither a hardware gate compilation nor a T-count claim. State preparation costs c_ψ query units, with serial depth d_ψ specified separately if needed. The depth restriction is $d_\psi + 4n + 1 \leq D$; the last call performs readout. In the numerical

model we conservatively set $d_\psi = c_\psi$. Thus all methods within a given problem obey the same full serial-query cap.

Given an accepted state, the success flag of a block-encoding of B is a Bernoulli variable of mean μ_P. For total normalized Gram error E, allocate $0 < b < E$ to bias and $E - b$ to sampling. With

$$M(b) = \left\lceil \frac{\log(2/\delta)}{2(E-b)^2} \right\rceil \tag{2}$$

independent accepted copies, Hoeffding's inequality yields $|\widehat{\mu} - e| \le E$ with probability at least $1-\delta$. Our restart protocol discards every failed whole-filter attempt, prepares the input again, and uses no amplitude amplification or intermediate recycling. Its worst expected query cost is exactly

$$J(P, n, b) = M(b) \left[\frac{c_\psi + 4n}{\kappa^2 S(P)} + 1 \right]. \tag{3}$$

Independence of fresh attempts justifies the geometric waiting time and the identical conditional distribution of accepted readouts.

The right minimum singular value is $\sigma_R(A) = \min_{\|v\|=1} \|Av\| = \alpha\sqrt{e}$. Clipping readout to $[0, 1]$ and taking a square root gives error at most $\alpha\sqrt{E}$. For wide matrices this right minimum is zero and need not equal the smallest entry of the thin SVD. Prior acquisition cost is an additional common input cost, not included in the optimization; acquiring a tighter prior adaptively is outside the model.

The no-computation baseline. Returning $u/2$ already estimates e to $u/2$; returning $\alpha\sqrt{u}/2$ estimates σ_R to $\alpha\sqrt{u}/2$. These strategies must be preferred whenever the requested error allows them. Our main numerical tests impose $u > 4E$, so neither midpoint solves even the singular-value task with tolerance $\alpha\sqrt{E}$. The unfiltered mean has worst bias $1 - a$ and is feasible only if its allocated bias is at least that large. These observations concern this mean estimator, not every possible unfiltered quantum algorithm.

3 Two-point characterization and endpoint reduction

Proposition 1 (Uniform mean-bias condition)**.** *An amplitude P nonzero on $[0, u]$ is admissible at $b > 0$ if and only if for every $0 \le e \le u$ and $e + b < x \le 1$,*

$$(1-a)(x-e-b)P(x)^2 \le abP(e)^2. \tag{4}$$

Proof. Necessity follows by placing mass a at e and $1-a$ at x and cross-multiplying the positive posterior denominator. For sufficiency, let actual target mass be $p_0 \ge a$. Equation (4) implies $(x-e-b)P(x)^2 \le abP(e)^2/(1-a)$ when $x - e > b$; if $x - e \le b$, the left side is nonpositive. Averaging over the non-target conditional distribution gives

$$\sum_{i>0} p_i(\lambda_i - e - b)P(\lambda_i)^2 \le \frac{(1-p_0)ab}{1-a} P(e)^2 \le p_0 bP(e)^2.$$

This is exactly the bias inequality after cross-multiplication. □

For a fixed e and exact mass a, the same averaging argument gives the exact worst mean bias

$$\max_{e \le x \le 1} \frac{(1-a)(x-e)P(x)^2}{aP(e)^2 + (1-a)P(x)^2}. \tag{5}$$

Thus finite examples can attain, rather than merely suggest, the worst-case bounds. This is a simple linear-fractional extremization; the general principle is classical [10].

Theorem 2 (Controlled endpoint nulling)**.** *Let $u + b < 1$ and let P be continuous and admissible at b, with $|P| \le 1$. Then $Q(x) = (1 - x)P(x)$ is admissible, $|Q| \le 1$, $Q(1) = 0$, and*

$$S(Q) \ge (1 - u)(1 - u - b)S(P). \tag{6}$$

If P has degree at most n, Q has degree at most $n + 1$ in λ.

Proof. For $x \ge e$, $0 \le 1 - x \le 1 - e$. Multiplying (4) by $(1 - e)^2$ therefore proves the condition for Q. Let $m = \min_{[0,u]} P^2 > 0$, attained at e_*. Every permitted input gives $Z_Q \ge a(1 - u)^2 m$. For P, use the allowed witness with masses $a, 1 - a$ at $e_*, 1$. Equation (4) bounds its endpoint contribution, giving

$$S(P) \le am + (1 - a)P(1)^2 \le am\frac{1 - e_*}{1 - e_* - b} \le am\frac{1 - u}{1 - u - b}.$$

Combining the two inequalities proves (6). □

Corollary 3 (Cost of the restriction)**.** *For the same b, M, κ, transforming a degree-n filter to an endpoint-nulling degree-$(n + 1)$ filter increases the worst expected cost by at most*

$$\frac{c_\psi + 4(n + 1)}{c_\psi + 4n}\frac{1}{(1 - u)(1 - u - b)}. \tag{7}$$

The additional degree must fit the depth cap. Consequently the restriction is quantitatively controlled, but is not cost-free at a tight cap.

Theorem 4 (Exact restart probability after endpoint nulling)**.** *If P is continuous, $P(1) = 0$ and $P > 0$ on $[0, u]$, let $\tau(P) = \min_{[0,u]} P$. Then*

$$S(P) = a\tau(P)^2. \tag{8}$$

Proof. The target component gives the lower bound. Take a minimizer e_* of P on $[0, u]$ and put mass a there and the remaining mass at 1. This witness attains the bound exactly. □

The factor in Theorem 2 is the main structural reason to use endpoint-nulling polynomials. It gives a meaningful comparison to the larger real-polynomial class while making acceptance optimization exact. It does not compare to amplitude-amplified or adaptive filtering protocols.

4 The polynomial selection problem

For $n \ge 1$ let $\tau_n^*(a, u, b)$ be the optimum of

$$\begin{aligned} &\text{maximize } \ \tau \quad \text{over real } P \in \mathcal{P}_n,\ \tau \ge 0, \\ &\text{subject to } \ |P(x)| \le 1 \quad (0 \le x \le 1), \\ &\qquad P(1) = 0, \quad P(e) \ge \tau \quad (0 \le e \le u), \\ &\qquad \pm P(x) \le \sqrt{\frac{ab}{(1 - a)(x - e - b)}}P(e) \quad (0 \le e \le u,\ e + b < x \le 1). \end{aligned} \tag{9}$$

All constraints are linear in the polynomial coefficients and τ. The square-root coefficient is fixed by the supplied prior and error. Signed response outside the target interval is allowed. For positive τ, Proposition 1 makes these constraints equivalent to admissibility in the endpoint-nulling class. Compactness of bounded degree-n polynomials gives attainment; an optimum of zero means no positive uniformly accepting design is supplied at that degree.

Theorem 5 (Restricted optimum and certified selection)**.** *For fixed n, b, the smallest worst expected cost in this class is*

$$J_n^*(b) = M(b)\left[\frac{c_\psi + 4n}{\kappa^2 a(\tau_n^*)^2} + 1\right]. \tag{10}$$

Suppose a certified feasible polynomial gives $\ell_{n,b} \le \tau_n^*$ and a valid relaxation gives $\tau_n^* \le U_{n,b}$. Define J^{up} by replacing τ_n^* with $\ell_{n,b}$, and J^{lo} by replacing it with $U_{n,b}$. For any finite allocation set $\mathcal{B} \subset (0, E)$, selecting the feasible (n, b) with smallest J^{up} gives

$$\frac{J_{\mathrm{selected}}}{\min_{n\le N,\ b\in\mathcal{B}} J_n^*(b)} \le \frac{\min J^{\mathrm{up}}}{\min J^{\mathrm{lo}}}. \tag{11}$$

This is an a posteriori approximation factor for the stated degree, allocation, restart and readout model.

Proof. Equation (10) follows from Theorem 4 and monotonicity of cost in τ. Every valid upper/lower transmission bound gives the corresponding reverse cost bound. Taking minima proves (11). A comparison family with its own continuous certificate may supply the selected upper bound as well. □

With fixed M, b, the cost of candidate i is affine in preparation cost, $J_i(c) = M[(c+4n_i)/s_i + 1]$, where s_i is its worst implemented success. Two candidates with $s_j > s_i$ and $n_j > n_i$ exchange order at

$$c_* = \frac{4(n_j s_i - n_i s_j)}{s_j - s_i}. \tag{12}$$

Transmission intervals turn this point into a certified transition interval: an upper cost for one candidate below a lower cost for the other proves its ordering. This explains why maximizing acceptance alone and minimizing degree alone need not select the same response.

5 A filter-independent prior-width limitation

The paired requirement also limits what any fixed response can achieve, even without a polynomial or degree restriction. This provides a useful early rejection test for proposed acceptance budgets. Set $A_b = ab/(1-a)$ and $\Delta_0 = b/(1-a)$. For $z \ge 0$ define

$$F_b(z) = \max\left(\{0\} \cup \left\{ n \log\frac{z/n - b}{A_b} : n \in \mathbb{N}_{>0},\ z/n > \Delta_0 \right\}\right). \tag{13}$$

Only finitely many integers contribute.

Proposition 6 (Exact unrestricted target transmission)**.** *For measurable weights $0 \le w \le 1$, positive on $[0, u]$ and obeying the paired mean-bias condition, one has*

$$\sup_w \inf_{0\le e\le u} w(e) = e^{-F_b(u)}. \tag{14}$$

In particular, some permitted pure-target input has expected restart overhead at least $e^{F_b(u)}$ for any one fixed filter.

Proof. Applying the ratio form of (4) along a partition of $[0, u]$ with step lengths $d_j > \Delta_0$ gives $w(u) \le \prod_j A_b/(d_j - b)$. For a fixed number n of steps, concavity of $\log(d - b)$ makes the equally spaced partition strongest, giving the upper bound in (14). For completeness, allowing arbitrary step lengths with total length at most z gives exactly (13): leftover length can be added to a step, and equal spacing maximizes its logarithmic attenuation. Concatenating partitions shows that F_b is superadditive; it is also nondecreasing. Thus $w_*(x) = e^{-F_b(x)}$ satisfies $w_*(e+d)/w_*(e) \le A_b/(d-b)$ when $d > \Delta_0$. For $b < d \le \Delta_0$ the required ratio bound is at least one and monotonicity suffices. Therefore w_* is feasible on the entire support and attains the minimum at u, proving equality. A pure eigenstate at u has acceptance $w(u)$, proving the restart statement. □

The large-width exponential rate is r_a/b, where $r_a = W_0((1-a)/(ae))$ and W_0 is the nonnegative Lambert branch. Indeed, maximizing $\log((d - b)/A_b)/d$ gives $d_* = b(1 + 1/r_a)$. The exponential weight $\exp(-r_a x/b)$ satisfies the pair condition and provides a familiar imaginary-time comparison. Equation (14) is a function-level bound, not a general quantum-query lower bound. It does not apply to algorithms that adapt the filter after learning more about the minimum, use a different estimator, or amplify/recycle successes. It is also not a proof that no polynomial can attain the same value: nonanalyticity of the displayed optimizer alone would not establish such a statement.

6 Continuous and phase-realization certificates

We use Chebyshev coefficients $P(x) = \sum_{k=0}^{n} c_k T_k(2x - 1)$. Finite-grid linear programming and a few exchange rounds produce candidates, but no grid result is called a continuous certificate. All saved decimal coefficients are interpreted as exact rationals, with $P(1) = 0$ enforced algebraically. Bernstein coefficients and outward integer subdivision at 220-bit fixed-point precision enclose the polynomial on whole intervals [11]. They certify boundedness, positive target transmission and (4) on the full $(e, x) \in [0, u] \times [0, 1]$ rectangle. Nonphysical or already-small separations are harmless. Exact normalization by the computed global bound is applied once to the composite polynomial.

For an upper transmission bound, pair square roots in the grid relaxation are rounded upward to rational numbers, which weakens the constraints. Floating solver multipliers are converted to nonnegative rational weights. If y is this vector, r is its exact stationarity residual, and d is the objective vector, weak duality gives

$$d^T z \le y^T h + \sum_j |r_j| B_j, \qquad r = d - A^T y, \tag{15}$$

because the true coefficient bounds are $|c_0| \le 1$, $|c_k| \le 2$ for $k > 0$, and $0 \le \tau \le 1$. These Chebyshev bounds follow directly from the cosine-integral formula for their coefficients. The stationarity correction is kept even when numerically small. Thus an imperfect solver dual is not silently treated as an exact optimum.

The entire even polynomial $\kappa P(s^2)$ is synthesized once with symmetric QSP using the `pyqsp` implementation of a Newton solver [13]. The signal convention is

$$W(s) = \begin{pmatrix} s & i\sqrt{1-s^2} \\ i\sqrt{1-s^2} & s \end{pmatrix}, \qquad U_\Phi = e^{i\phi_0 Z} \prod_{j=1}^{2n} [W(s) e^{i\phi_j Z}].$$

The desired response is $\mathrm{Im}(U_\Phi)_{00}$, obtained using the success block of $(-iU_\Phi + iU_\Phi^\dagger)/2$. Controlled branches implemented serially justify the $4n$ accounting. Saved phase decimals define exact real angles. Rational Taylor remainder enclosures for each sine/cosine, an interval polynomial recurrence

for the top row of U_Φ, and Bernstein subdivision give a uniform response error η on the whole singular-value interval. This is stronger than a floating grid agreement, while remaining a certificate for the ideal signal oracle rather than a hardware error bound.

Proposition 7 (Accepted-state error transfer)**.** *Let the ideal filter success probability be at least $s > 0$, and let the implemented success block differ by operator norm at most $\eta < \sqrt{s}$. Then its success probability is at least $(\sqrt{s} - \eta)^2$ and its accepted Gram mean differs from the ideal mean by at most $2\eta/\sqrt{s}$.*

Proof. The unnormalized state error is at most η. Reverse triangle inequality proves the success bound. Normalizing changes vector error by at most a factor $2/\sqrt{s}$; pure-state trace distance is no greater than vector distance, and the expectation of $0 \le G \le I$ changes by at most trace distance. □

Our polynomial certificate reserves $0.01b$ of the bias allocation for this transfer, and every synthesized selected response must satisfy it. Encoding errors would also have to be included in η before applying the proposition. Applying a safety scale at every scalar iteration would instead cause an avoidable exponential penalty; synthesizing the whole polynomial once avoids that particular artifact.

7 Numerical study

We use $\delta = 0.05$, $\kappa = 0.95$ and the finite allocation grid $b/E \in \{0.4, 0.6, 0.8\}$. Eight cases use $a \in \{0.3, 0.6\}$, $u \in \{0.125, 0.2\}$, $E = 0.025$, $c_\psi \in \{1, 32\}$ and $N = 20$. Two further cases use $a = 0.6$, $u \in \{0.25, 0.4\}$, $E = 0.05$, $c_\psi = 1$ and $N = 12$. The full serial-query caps are $c_\psi + 4N + 1$ for each problem. All cases satisfy $u > 4E$, excluding the prior-midpoint solution. The parameter c_ψ is a sensitivity parameter of the cost model; we do not claim that preparing the simple matrix-audit input costs 32 queries on actual hardware.

Candidates and certificates. The comparison families are endpoint-nulling power $(1 - x)^n$, quartic-power $(1-x^2)^t$, endpoint-nulled cosine-Taylor products $(1-x)(1-0.72x^2)^t$, and the classical low-pass Chebyshev polynomial

$$(1 - x)\frac{T_{n-1}((2x - 1 - v)/(1 - v))}{T_{n-1}(-(1 + v)/(1 - v))}, \quad v \in \{.04, .08, .16, .28, .42, .60, .78\}.$$

The degree is always measured in $x = \lambda$ and then converted to signal queries identically. Normalization, target positivity and bias are certified for every reported candidate. Optimized candidates use initial 17-target/65-tail/129-support grids, with up to three exchange rounds against denser grids. The LP design bias is $0.98b$; the continuous polynomial certificate uses $0.99b$, leaving $0.01b$ for realization error. The verifier can reject a gridded optimum, so the selected design is the best continuously certified candidate among those attempted; its optimality gap remains controlled by the independent dual lower bound.

The selections have certified cost factors between 1.0372 and 1.2394 relative to the restricted optimum lower bound. Most selected responses come from the coupled LP, but the $a = 0.6, u = 0.25, E = 0.05$ case selects the tested Chebyshev response instead. Thus the implementation does not automatically prefer the newly optimized family. Power and Chebyshev candidates are certified in all ten cases. No quartic-power or cosine-Taylor candidate passed the requirements within the searched degree/parameter grid in these harder cases; this is recorded as absence of a certified

a	u	E	c_ψ	n	Response	b	J^{up}	Gap
0.30	0.125	0.025	1	8	LP	0.015	1.79e+08	9.5%
0.30	0.125	0.025	32	8	LP	0.015	3.47e+08	9.5%
0.30	0.200	0.025	1	8	LP	0.020	1.71e+09	23.9%
0.30	0.200	0.025	32	8	LP	0.020	3.31e+09	16.2%
0.60	0.125	0.025	1	4	LP	0.015	3.27e+06	21.4%
0.60	0.125	0.025	32	4	LP	0.015	9.19e+06	3.7%
0.60	0.200	0.025	1	4	LP	0.015	1.02e+07	7.5%
0.60	0.200	0.025	32	5	LP	0.015	2.72e+07	5.6%
0.60	0.250	0.050	1	2	Cheb.	0.030	5.1e+05	4.9%
0.60	0.400	0.050	1	3	LP	0.030	1.98e+06	7.4%

Table 1: Prior-only selections. n is Gram-polynomial degree. "LP" is a certified optimized polynomial; "Cheb." is a certified member of the tested Chebyshev grid. The gap is $100(J^{\mathrm{up}}/\min J^{\mathrm{lo}} - 1)$, relative to the endpoint-nulling class, degree cap and finite allocation grid, not to all quantum estimators.

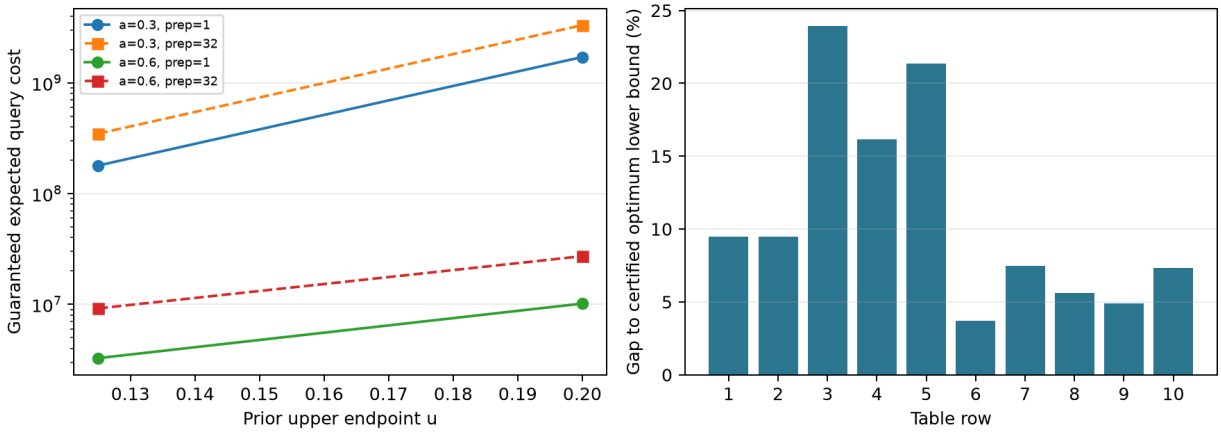


Figure 1: Guaranteed expected query costs and a posteriori gaps for the main cases. Different preparation costs are separate specified problems; all candidates within each problem share its cap and oracle accounting.

candidate, not a proof of impossibility for all tunings. Relative to the selected upper costs, the tested power upper costs range from approximately 1.44 to 4.36 times as large. These are comparisons of guaranteed costs in the common model, not empirical runtime speedups.

A preparation-cost-dependent switch. For $a = 0.6, u = 0.2, E = 0.025$ and $b = 0.015$, preparation cost one selects degree four, whereas preparation cost 32 selects degree five. Both use the same accepted-copy sample count. The latter response has greater guaranteed acceptance, enough to compensate for the additional queries when input preparation is expensive. For the ideal polynomial responses, exact transmission intervals give the crossover interval reported in the supplementary numerical summary; the central displayed guaranteed-cost crossover is approximately 12.34 preparation-query units. This comparison concerns two fixed certified polynomials; it is not a claim that these are the only optimal candidates at every intermediate preparation cost.

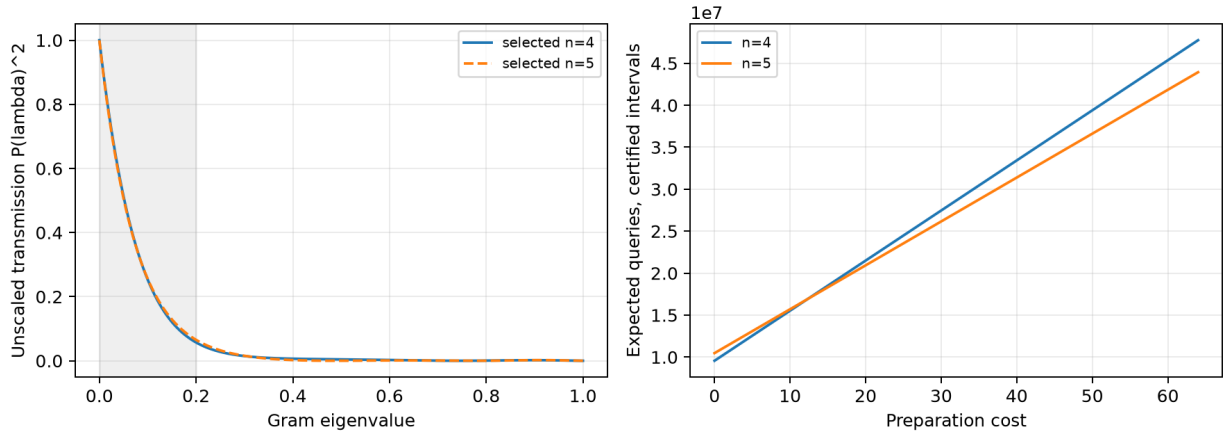


Figure 2: Two selected responses for the same prior and bias allocation. The shaded spectral interval is the supplied prior. Their certified cost lines exchange order as preparation cost increases; transmission enclosures produce the narrow bands.

The price of prior width. Figure 3 compares the certified target transmissions with the unrestricted bound of Proposition 6. None exceeds the function-level limit. For example, at $a = 0.3$ widening the prior from 0.125 to 0.2 at $E = 0.025, c_\psi = 1$ increases the selected guaranteed cost from about 1.79×10^8 to 1.71×10^9 queries. These large absolute costs do not demonstrate practicality; they illustrate the limitations of this robust filtered-mean protocol.

Phase and matrix audits. All ten selected whole polynomials were synthesized successfully. The largest certified continuous amplitude error of the saved phase responses is 3.63×10^{-16}. Each satisfies the reserved mean-bias allowance; the certified worst-success correction changes the cost by less than a factor $1 + 1.4 \times 10^{-14}$. These statements apply to the ideal signal convention, not to hardware rotation synthesis or noisy block-encodings.

For each selection we also construct an eight-dimensional positive matrix B with a minimum in the supplied interval and use the fixed computational input $|0\rangle$. A Householder rotation in the synthetic generator controls its overlap promise; the estimator never receives that rotation or the true eigenvalues for its design. One case of each pair includes an excitation only 10^{-7} above the minimum. We apply every QSP phase to the full 16×16 signal matrix and explicitly form the 32×32 extraction unitary. The maximum response-matrix error against the polynomial target is 5.55×10^{-16} and the maximum unitarity residual is 6.33×10^{-15}. This is an exact-oracle matrix simulation, not physical-gate compilation.

Across 1,000 held-out scalar spectral inputs (seed 20261027), the guaranteed bias and success bounds held. A further 2,000 aggregate Bernoulli/negative-binomial replicates for the full matrix examples (seed 20261028) had no confidence failures at the requested error. The confidence guarantee comes from the theorem; observing zero finite sample failures is only a consistency check. All coefficients, phases, input matrices, sample settings and exact certificates are included.

8 Discussion and limitations

The selection rule answers a concrete question with supplied prior information: which real polynomial degree, response and finite bias allocation minimize a guaranteed worst restart cost? The

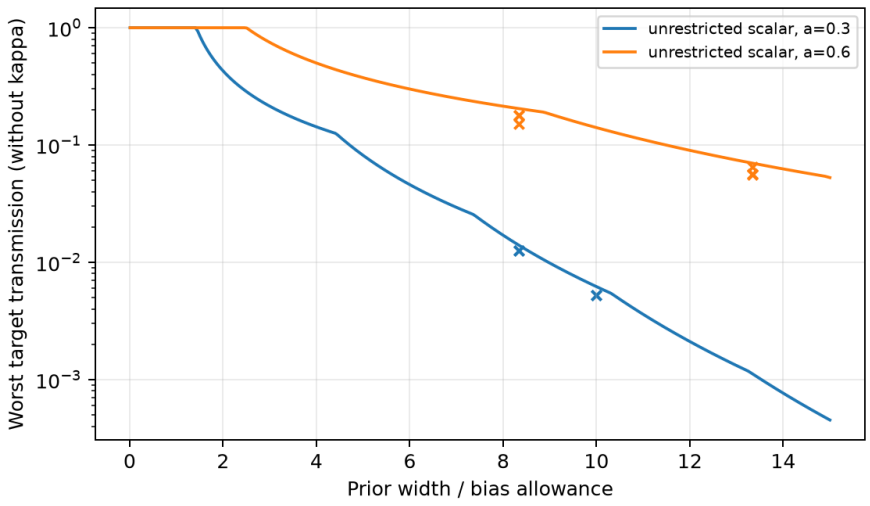


Figure 3: The exact unrestricted target-transmission limit as a function of prior width divided by bias allowance, and certified polynomial transmissions (crosses). The margin κ is excluded from both quantities here.

endpoint reduction makes the restriction explicit and controls the price of using the class in which the convex objective has an exact operational meaning. The dual/continuous certificates distinguish proximity to a restricted optimum from improvement over a small list of named filters.

The estimator and restart protocol are substantial restrictions. Amplitude amplification, coherent recycling, virtual filtering, phase-estimation-based readout and adaptive prior acquisition can change the resource dependence. Modern QET-U and other ground-energy estimators address different information/readout regimes [14, 15]; our results are not lower bounds on those algorithms. A supplied overlap bound can itself be costly to obtain. Gate-level depth, error correction, chemistry block-encodings and hardware noise are not modeled here. Optimization over continuous bias allocations is not certified by a finite allocation grid. Numerical near-optimality factors must always be read with this scope attached.

The worst-case model can be conservative for a particular spectrum, and the total costs can be large when prior width is large compared with the bias allowance. That behavior is informative: filter flexibility alone does not remove the price of weak prior information. The theory does not claim a new family of optimal QSVT polynomials or a general quantum advantage. Its contribution is the structural reduction and a reproducible, certified answer for the specified robust selection problem.

9 Conclusion

We formulated prior-only gap-free filter selection for a spectral-mean estimator with a full restart/readout query accounting. A controlled endpoint-nulling reduction connects general real responses to a class whose worst acceptance is exactly characterized. This yields a convex polynomial formulation with rigorous lower and upper cost certificates, and selected phase responses can be checked on continuous intervals. The framework identifies when a change in preparation cost or prior information warrants a different response, while exposing the regimes where a fixed-filter mean estimator remains expensive.

A Exact arithmetic and certificate construction

For $p(x) = \sum_{j=0}^{n} a_j x^j$ on $[0, L]$, its degree-n Bernstein coefficients are

$$b_i = \sum_{j=0}^{i} a_j L^j \frac{\binom{i}{j}}{\binom{n}{j}}.$$

Because the Bernstein basis is nonnegative and sums to one, the polynomial lies between the smallest and largest coefficients. Midpoint subdivision uses de Casteljau averaging. The verifier rounds each lower endpoint downward and each upper endpoint upward to integer multiples of 2^{-220}. Induction on subdivision proves containment. All saved coefficient and phase decimals are exact rational inputs; the floating optimization outputs themselves are not exact certificates.

To certify the coupled residual, write $w = P^2$ and $A_b = ab/(1-a)$. On a box $e \in [e_0, e_1]$, $x \in [x_0, x_1]$, use $w(x) \geq 0$ to bound

$$(x - e - b)w(x) - A_b w(e) \leq (x - e_0 - b)w(x) - A_b w(e).$$

The verifier encloses $xw(x)$ and $w(x)$ in the same elevated Bernstein basis, subtracts $(e_0 + b)w(x)$, and subtracts a lower bound for $A_b w(e)$. Boxes with $x_1 \leq e_0 + b$ need no further check. If a box cannot be resolved within the recorded node/depth cap, the candidate is rejected as uncertified, even if dense-grid tests pass. This is a certificate method, not a guaranteed optimizer or guaranteed-termination theorem.

For the phase-response recurrence write the top row as $(P(s), i\sqrt{1-s^2}Q(s))$. One further signal/phase step gives

$$P_{\text{new}} = e^{i\phi}[sP - (1-s^2)Q], \qquad Q_{\text{new}} = e^{-i\phi}[P + sQ].$$

For even signal degree the diagonal polynomial has even parity, so its imaginary part is a polynomial in s^2. Rational Taylor remainders for $\sin\phi, \cos\phi$ enclose each complex multiplication. The resulting error polynomial in $\lambda = s^2$ is bounded with the same Bernstein method. The stated phase bound covers a continuous interval, not only the grid used for the separate numerical matrix check.

At $\delta = 0.05$, the implementation obtains an upper rational bound for $\log 40 = \log 5 + 3\log 2$ using

$$\log x = 2\sum_{j=0}^{m-1} \frac{z^{2j+1}}{2j+1} + R_m, \quad z = \frac{x-1}{x+1}, \quad 0 \leq R_m \leq \frac{2z^{2m+1}}{(2m+1)(1-z^2)}.$$

It then rounds the rational shot requirement upward. Thus the integer sample count is not based on an unchecked floating rounding at a ceiling.

B Formal verification scope

`lean/Selection.lean` was checked locally with Lean 4.35.0 and Mathlib v4.35.0 by `lake build Selection`. `lake env lean SelectionAudit.lean` also succeeded. The five checked scalar lemmas concern: the two-point cross-multiplication equivalence; sufficiency for a normalized finite outside mixture with target mass exactly a; preservation of the pair inequality under ordered attenuation factors; the algebraic endpoint acceptance factor from its two input bounds; and weak duality with a stationarity residual correction. The audit reports only `propext`, `Classical.choice`, and `Quot.sound`. No `sorry`, `admit`, custom axiom, or `native_decide` is used in those proofs.

The proof for actual target mass $p_0 \geq a$, the optimization/function-space arguments, the prior-width envelope, Bernstein algorithm, phase-realization enclosures, matrix reduction, query implementation, and concentration theorem are hand proofs or computational certificates, not fully formalized Lean theorems. A workflow for the same Lean target is provided, but no remote CI run is claimed; the reported success is local. The supplementary verification record includes source hashes and logs.

C Reproducibility and scope of computation

The source package includes these entry points:

- `filter_selection.py`: design and continuous certification;
- `selection_qsp.py`: phase synthesis and response enclosure;
- `selection_matrix_audit.py`: full block-matrix audit;
- `selection_figures.py`: tables, figures and summary;
- `check_selection_certificates.py`: exact arithmetic controls.

Their shared arithmetic module is `selection_certificates.py`. The design entry point records source hashes, base Git revision when available, rational coefficients, exact dual bounds, certificate diagnostics, parameters and software versions. The QSP JSON links to its design input by hash, and the matrix JSON links to both design and phase inputs. Exact-cost rationals are provided along with rounded displays. A complete source/hash manifest is included.

The main experiments use a small synthetic prior grid. Named baseline parameters are searched on a finite documented grid, so comparisons are to those certified candidates, not to every possible tuning of a named family. Some families fail the degree/bias/certificate requirements; no performance value is invented for them. Every main case has $u > 4E$. Earlier diagnostic cases that violated this nontriviality condition were excluded from the reported study; they are recorded as a rejected evaluation design in the research notes.

Author information is intentionally omitted in this review copy. AI assistance was used for candidate derivations, drafting and numerical implementation; the verification scopes above identify the actual checks rather than equating generation with verification. The package has not been uploaded or submitted to a preprint server.

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